LAB 7 REPORT: Fuzzing for Input Validation Bugs

Course: Software Testing & Automation

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Repository: L6_L7_13_pet_walking_service

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1. Objective

To detect, analyze, and eliminate input validation bugs in a pet-walking service application by applying property-based fuzz testing using Python's Hypothesis library.

2. Tools & Environment

Item	Details
Operating System	Linux (Ubuntu)
Programming Language	Python 3.12
Testing Frameworks	pytest (v9.1.1), hypothesis (v6.170.15)

3. Execution Steps & Screenshots

Step 1: Initial Application Run

Command: `python3 processor.py`

Description: Ran processor.py with the sample inputs 2-30-YES, 45/3 and CALM. The schedule step printed Walk: 2 pets, 30 mins - Total $45.00 and the temperament step printed Surcharge: $ 0.0. The pace step printed Pace (min/pet): None for the valid input 45/3, so a defect was visible before any fuzzing.

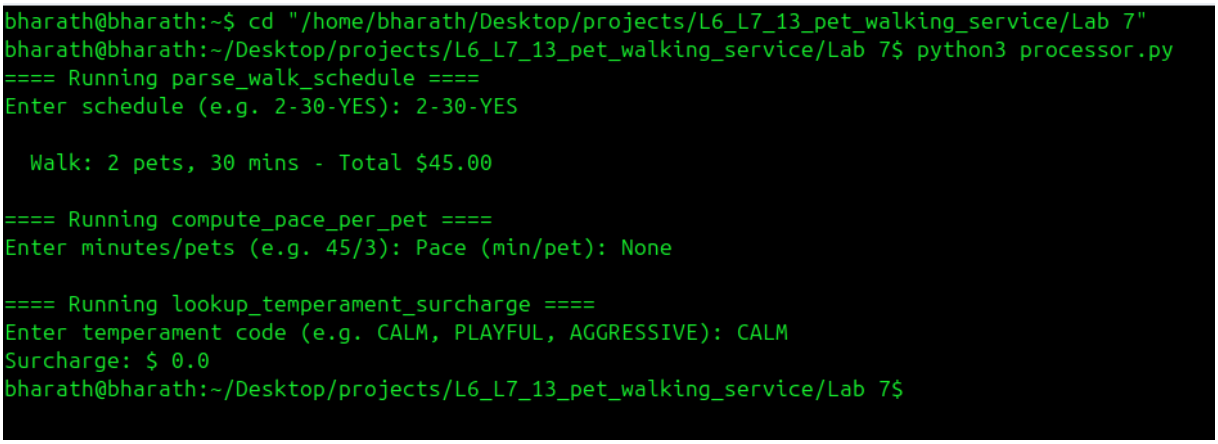


Figure 1: *processor.py run with valid sample inputs. The pace step prints None for 45/3.*

Step 2: Fuzz Testing with pytest and Hypothesis

Command: `pytest test_processor.py`

Description: Ran the Hypothesis-based test suite in test_processor.py against processor.py. pytest reported 3 passed in 0.21s.

```
bharath@bharath:~/Desktop/projects/L6_L7_13_pet_walking_service/Lab 7$ pytest test_processor.py

platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/bharath/Desktop/projects/L6_L7_13_pet_walking_service/Lab 7
plugins: platformdirs-4.13.7, hypothesis-6.170.15, asyncio-1.4.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function


test_processor.py ...

============================================================================== 3 passed in 0.21s ===============================
bharath@bharath:~/Desktop/projects/L6_L7_13_pet_walking_service/Lab 7$
```

Figure 2: *pytest test_processor.py output: 3 passed in 0.21s.*

Step 3: Re-run after git checkout

Command: `git checkout processor.py; python3 processor.py`

Description: Restored processor.py with git checkout processor.py and ran it again with the same sample inputs. The pace step now prints Pace (min/pet): 15.0 for 45/3, and the schedule and surcharge outputs are unchanged.

```
bharath@bharath:~/Desktop/projects/L6_L7_13_pet_walking_service/Lab 7$ git checkout processor.py
Updated 1 path from the index
bharath@bharath:~/Desktop/projects/L6_L7_13_pet_walking_service/Lab 7$ python3 processor.py
==== Running parse_walk_schedule ====
Enter schedule (e.g. 2-30-YES): 2-30-YES
Walk: 2 pets, 30 mins - Total $45.00

==== Running compute_pace_per_pet ====
Enter minutes/pets (e.g. 45/3): 45/3
Pace (min/pet): 15.0

==== Running lookup_temperament_surcharge ====
Enter temperament code (e.g. CALM, PLAYFUL, AGGRESSIVE): CALM
Surcharge: $ 0.0
bharath@bharath:~/Desktop/projects/L6_L7_13_pet_walking_service/Lab 7$
```

Figure 3: *processor.py after git checkout processor.py. compute_pace_per_pet returns 15.0 for 45/3.*

Step 4: Final Run of the Fuzz Test Suite

Command: `pytest test_processor.py`

Description: Ran the same test suite again as the final check. pytest reported 3 passed in 0.21s.

```
bharath@bharath:~/Desktop/projects/L6_L7_13_pet_walking_service/Lab 7$ pytest test_processor.py

platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0
rootdir: /home/bharath/Desktop/projects/L6_L7_13_pet_walking_service/Lab 7
plugins: platformdirs-4.13.7, hypothesis-6.170.15, asyncio-1.4.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function


test_processor.py ...

============================================================================== 3 passed in 0.21s ===============================
bharath@bharath:~/Desktop/projects/L6_L7_13_pet_walking_service/Lab 7$
```

Figure 4: *Second pytest test_processor.py run: 3 passed in 0.21s.*

4. Summary of Bugs Found & Fixes Applied

Function	Input Exposing Bug	Exception	Fix Implemented
`parse_walk_schedule`	`s=""`, `s=None`	ValueError, AttributeError	Added string type checks and token count verification (len == 3).
`compute_pace_per_pet`	`s=""`, `s="/"`, `s="30/0"`	ValueError, ZeroDivisionError	Added division-by-zero checks (pets <= 0), delimiter checks, and filtered NaN/Inf.
`lookup_temperament_surcharge`	`s=""`, `s="XYZ"`	KeyError	Used dictionary .get(prefix, 0.0) for safe fallback.

5. Conclusion

Property-based fuzz testing with Hypothesis identified boundary bugs (empty inputs, null values, and divide-by-zero operations) that traditional unit testing overlooked. The codebase was hardened using defensive programming practices to ensure graceful error handling.